DAY 2 – Networking Basics (IP, DNS, Ports)

Task: Find IP Address of 3 Websites

Introduction

Every website on the internet is identified by an IP address. We can find the IP address of a website using networking tools like ping and nslookup.

Method 1:

Using ping Command:

- ping example.com
- This sends packets to the website and shows the IP address in brackets.
- Example: ping flipkart.com → Pinging flipkart.com [163.53.76.86]

Method 2:

Using nslookup Command:

- nslookup example.com
- This queries DNS servers to get the IP address of a domain.
- Example: nslookup myntra.com → Address: 184.28.109.49

Steps to Perform the Task:

1. Open Terminal
2. Use ping command for 3 websites (e.g: flipkart.com, myntra.com, instagram.com)
3. Note the IP address shown in brackets
4. Use nslookup command for the same websites
5. Record the IP addresses obtained

Example Results:

Website:flipkart.com

- Ping IP: 163.53.76.86
- Nslookup IP: 163.53.76.86

Website: myntra.com

- Ping IP: 184.28.109.49
- Nslookup IP: 184.28.109.49

Website: instagram.com

- Ping IP: 57.144.208.34
- Nslookup IP: 57.144.208.34

Key Differences

Feature	ping	nslookup
Purpose	Check connectivity	DNS query
Main Use	Network testing	Finding IP
Shows IP	Yes	Yes
Checks response time	Yes	No
DNS details	Limited	Detailed
SOC Usage	Basic check	DNS investigation

Conclusion:

Both ping and nslookup can be used to find the IP address of a website. nslookup is more reliable for DNS queries, while ping also checks connectivity.

DNS (Domain Name System)

What is DNS?

DNS stands for Domain Name System. It is like a phonebook of the internet. It converts human-friendly domain names (like google.com) into IP addresses (like 142.251.43.142).

Why DNS is Important?

Computers understand IP addresses, not names. DNS helps users access websites using easy names instead of numbers.

How DNS Works (Step-by-Step):

1. You type a website name in your browser (e.g., google.com)
2. Your computer sends a request to a DNS server
3. The DNS server finds the IP address of that website
4. It returns the IP address to your computer
5. Your browser connects to that IP and loads the website

Key Components of DNS

- Domain Name: Name of the website (google.com)
- IP Address: Numeric address of the server
- DNS Server: System that translates names to IPs

Types of DNS Servers

- Recursive Resolver: Receives the request from user
- Root Server: First step in finding IP
- TLD Server: Handles domains like .com, .org
- Authoritative Server: Provides final IP address

Example:

Domain: google.com → IP: 142.251.43.142

Conclusion:

DNS makes the internet user-friendly by allowing us to use names instead of IP addresses

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Common Network Ports and Their Uses

Introduction:

Ports are communication endpoints used by computers to send and receive data over a network. Each port is associated with a specific service or protocol.

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3 Common Ports

Port Number	Protocol	Use
80	HTTP	Used for normal web browsing
443	HTTPS	Used for secure web browsing (encrypted)
21	FTP	Used for file transfer between systems

Explanation:

- Port 80 (HTTP): Used to access websites without encryption.
- Port 443 (HTTPS): Used for secure communication over the internet.
- Port 21 (FTP): Used to transfer files between client and server.

Conclusion: Understanding ports is important in networking and cybersecurity. SOC analysts monitor ports to detect suspicious activities.